

Email Spam Classifier: Minor Project Report

Authors: Srustisri Panda (sru-codes), Puja Rani Mishra (mishrapujarani11-web), Kajal Roul (kajalroul2007-ops)

Contact:

- Srustisri Panda: [LinkedIn](#)
- Puja Rani Mishra: mishrapujarani11@gmail.com, [LinkedIn](#)
- Kajal Roul: kajalroul2007@gmail.com, [LinkedIn](#)

Abstract

The Email Spam Classifier is an intelligent system designed to automatically categorize incoming emails as either spam or legitimate (ham). This project leverages Natural Language Processing (NLP) techniques and the Multinomial Naive Bayes algorithm to identify patterns and keywords commonly associated with fraudulent or unsolicited communications. By converting raw text into numerical data using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization and implementing stopword removal, the classifier achieves high efficiency and accuracy. The project provides a practical demonstration of text classification in a real-world application, offering a robust solution for managing unwanted emails. It includes a Colab-compatible Jupyter notebook for easy execution and exploration of the model, along with a professional README file detailing the project overview, technical stack, and functionality.

1. Introduction

Email remains a primary mode of communication, but its effectiveness is often hampered by the pervasive issue of spam. Unsolicited and often malicious emails not only clutter inboxes but also pose significant security risks. The Email Spam Classifier project addresses this challenge by developing an automated system to accurately distinguish

between legitimate emails and spam. This project serves as a practical application of Natural Language Processing (NLP) and machine learning, demonstrating how computational techniques can be used to manage and filter textual data effectively.

2. Project Objectives

The primary objectives of the Email Spam Classifier project are:

- To develop a highly accurate machine learning model capable of classifying emails as spam or ham.
- To implement essential Natural Language Processing (NLP) techniques, including text vectorization and stopwords removal.
- To utilize the Multinomial Naive Bayes algorithm for its efficiency and effectiveness in text classification tasks.
- To provide a clear and reproducible codebase, including a Colab-compatible Jupyter notebook for easy execution and understanding.
- To evaluate the model's performance using standard metrics such as Accuracy, Precision, Recall, and F1-Score.

3. Methodology

The project employs a **Multinomial Naive Bayes** algorithm, which is particularly well-suited for text classification due to its probabilistic nature and efficiency with discrete features (like word counts). The methodology involves the following steps:

- **Data Collection and Preprocessing:** A dataset of emails labeled as spam or ham is collected. Text preprocessing involves converting text to lowercase, removing punctuation, and eliminating common stopwords (e.g.,